THE UNIVERSITY of EDINBURGH
School of Physics and Astronomy

Hydrogen-Bond Percolation in Water from Ab Initio Molecular Dynamics (AIMD)

Peilin Yu
s24156743
Supervisor: Prof Graeme Ackland

Background and Aim

Water is structured by a fluctuating hydrogen-bond network rather than by fixed nearest neighbours. As temperature decreases and density increases, this network can shift from small disconnected clusters to a system-spanning connected structure.

This project aims to map the hydrogen-bond percolation line of water and plot it on the known phase diagram.

Theory

- ***Percolation Transition:***

When molecules form bonds and turn from isolated groups until one connected cluster stretches across the entire system - the percolation transition [1].

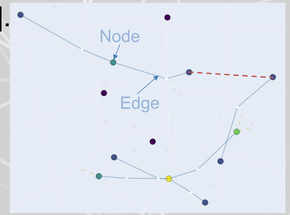

Fig. 1:Illustration of small clusters; if red dot line connect, system is percolated

We track this using two parameters:

1. The largest cluster fraction S = S_{max}/N: the fraction of molecules belonging to that system spanning cluster.
2. The mean coordination number <n_{HB}>: the average number of hydrogen bonds each molecule forms.

- ***Locating Transition Point:***

1. Susceptibility χ (the variance in largest cluster sizes)
2. Midpoint of sigmoidal curve

Result: Percolation line on phase diagram

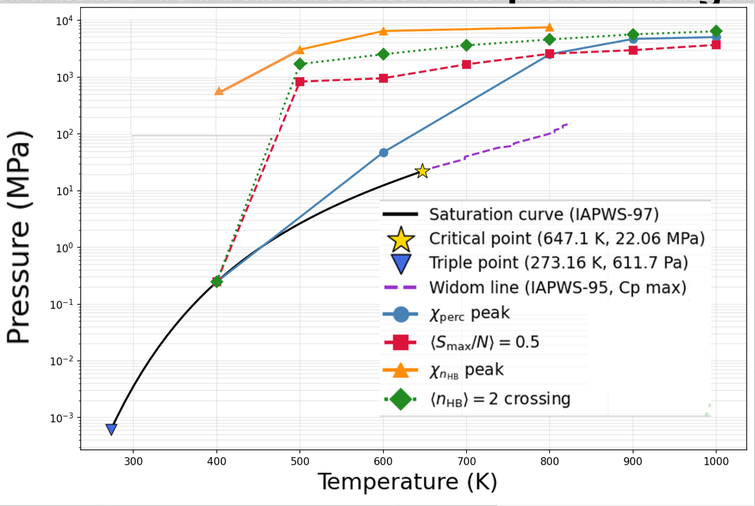

Fig. 2: Percolation line shown on the P–T phase diagram of water. Saturation line from IAPWS data [4].

- Percolation indicators map a smooth hydrogen-bond connectivity crossover across the phase diagram.
- In the P–T projection, the S_{max}/N=0.5, and <n_{HB}>=2 lines closely overlap.
- This suggests they capture the same structural crossover.
- The crossover remains meaningful even without a clear thermodynamic phase transition from critical point and triple point

Methods

AIMD trajectories of 100 water molecules [2] [3] were analysed over a range of temperatures and densities. Each simulation frame was converted into a hydrogen bond graph, with oxygen atoms as nodes(vertex) and edges(link) assigned when a hydrogen satisfied hydrogen bond distance (determined with the percolation cutoff point for the largest isochoric dataset). All analysis used a mean hydrogen bond cutoff r_c = 1.995±0.027Å.

Hydrogen-bond cutoff determination

Radial distribution function(RDF) based value fluctuated strongly across state points, the largest cluster fraction S_{max}/N normalised with its volume to eliminates the effect from increasing the volume, this showed clear, closely grouped peaks. This supported using a single representative cutoff, giving a more consistent definition of hydrogen-bond connectivity.

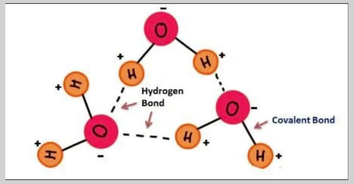

Fig 3: Schematic for water molecule and hydrogen bond

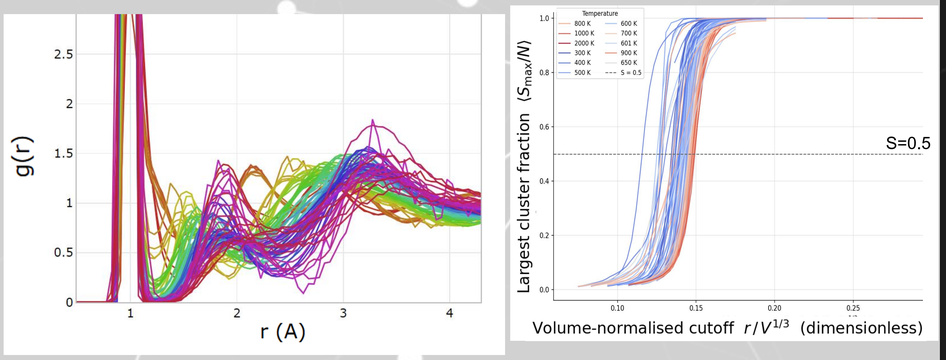

Fig 4: Left: O–H RDFs for all density and all temperature, showing strong fluctuation. Right: Largest-cluster fraction versus volume-normalised cutoff

Conclusions

Hydrogen-bond percolation in water defines a smooth connectivity crossover across the studied phase space, extending into the supercritical region.

A percolation-based cutoff also gives a more robust hydrogen-bond definition than the RDF criterion.

Future work should test whether this line correlates with thermodynamic, structural, or dynamical anomalies.

References : [1] Stauffer, D. & Aharony, A. *Introduction to Percolation Theory*. [2] Graeme Ackland unpublished data [3] Car, R. & Parrinello, M. (1985). [4] Wagner, W. & Pruß, A. (2002).